Econ 812 Recitation 2 Notes

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1 Steps to solving a Lagrangian DP problem

1. Set up Lagrangian, which has form of objective function + constraints*multiplier
2. Take FOCs
3. Combine FOCs to get Euler Equation (first order difference equation)
4. Solve for policy functions

2 Finite Horizon problem

$$\max_{c_t, k_{t+1}} \sum_{t=0}^{T} \beta^t u(c_t)$$

Subject to

$$c_t + k_{t+1} \leq z_t k_t^\alpha$$

$$k_{T+1} \geq 0$$

With given $k_0, \{z_t\}_{t=0}^T$

Step 1: Lagrangian

$$\mathscr{L} = \sum_{t=0}^{T} \beta^t [\log c_t - \lambda_t (c_t + k_{t+1} - z_t k_t^\alpha)] - \beta^T \mu_T (-k_{T+1})$$

Step 2: FOCs When $t \leq T$

$$c_t : \beta^t \frac{1}{c_t} - \beta^t \lambda_t = 0$$

When $t < T$

$$k_{t+1} : -\beta^t \lambda_t + \beta^{t+1} \lambda_{t+1} z_{t+1} \alpha k_{t+1}^{\alpha-1} = 0$$

When $t = T$

$$k_{T+1} : -\beta^T \lambda_T + \beta^T \mu_T = 0$$

Complementary slackness conditions:

$$\lambda_t : c_t + k_{t+1} = z_t k_t^\alpha$$

$$\mu_T : \beta^T \mu_T k_{T+1} = 0$$

Step 3: Combine FOCs to find Euler Equation Euler Equation:

$$\frac{c_{t+1}}{c_t} = \beta z_{t+1} \, \alpha \, k_{t+1}^{\alpha-1}$$

Step 4: Solve for policy functions We still have the budget constraint $c_t + k_{t+1} = z_t k_t^\alpha$

When $t = T$ we have that $c_T = z_T k_T^\alpha$

$$\frac{c_T}{c_{T-1}} = \beta z_T \, \alpha \, k_T^{\alpha-1}$$

$$\frac{z_T k_T^\alpha}{c_{T-1}} = \beta z_T \, \alpha \, k_T^{\alpha-1}$$

Now we can solve for the decision to consume in the previous period

$$c_{T-1} = k_T/(\alpha\beta)$$

For the T-2 period

$$\frac{c_{T-1}}{c_{T-2}} = \beta z_{T-1} \, \alpha \, k_{T-1}^{\alpha-1}$$

$$\frac{k_T/(\alpha\beta)}{c_{T-2}} = \beta z_{T-1} \, \alpha \, k_{T-1}^{\alpha-1}$$

$$c_{T-2} = \frac{k_T/\alpha\beta}{\beta z_{T-1} \, \alpha \, k_{T-1}^{\alpha-1}}$$

3 Infinite Horizon

$$\max_{c_t, k_{t+1}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

Subject to

$$c_t + k_{t+1} \leq z_t k_t^\alpha$$

With given $k_0, \{z_t\}_{t=0}^\infty$

Step 1: Lagrangian

$$\mathscr{L} = \sum_{t=0}^{\infty} \beta^t [\log c_t - \lambda_t (c_t + k_{t+1} - z_t k_t^{\alpha})]$$

Step 2: FOCs

$$c_t : \beta^t \frac{1}{c_t} - \beta^t \lambda_t = 0$$

$$k_{t+1} : -\beta^t \lambda_t + \beta^{t+1} \lambda_{t+1} z_{t+1} \alpha k_{t+1}^{\alpha-1} = 0$$

Complementary slackness conditon:

$$\lambda_t : c_t + k_{t+1} = z_t k_t^{\alpha}$$

TVC:

$$\lim_{t \to \infty} \beta^t \lambda_t k_{t+1} = 0$$

Step 3: Combine FOCs to find Euler Equation Euler Equation:

$$\frac{c_{t+1}}{c_t} = \beta z_{t+1} \, \alpha \, k_{t+1}^{\alpha-1}$$

Step 4: Solve for policy functions We still have the budget constraint $c_t + k_{t+1} = z_t k_t^{\alpha}$

$$\frac{z_{t+1} k_{t+1}^{\alpha} - k_{t+2}}{z_t k_t^{\alpha} - k_{t+1}} = \beta z_{t+1} \, \alpha \, k_{t+1}^{\alpha-1}$$

Expand terms

$$z_{t+1} k_{t+1}^{\alpha} - k_{t+2} = \beta z_{t+1} \, \alpha \, k_{t+1}^{\alpha-1} (z_t k_t^{\alpha}) - \beta z_{t+1} \, \alpha \, k_{t+1}^{\alpha}$$

Divide by $z_t k_t^{\alpha}$

$$z_{t+1} k_{t+1}^{\alpha} / z_t k_t^{\alpha} - k_{t+2} / z_t k_t^{\alpha} = \beta z_{t+1} \, \alpha \, k_{t+1}^{\alpha-1} - \beta z_{t+1} \, \alpha \, k_{t+1}^{\alpha} / z_t k_t^{\alpha}$$

Divide by $z_{t+1} k_{t+1}^{\alpha-1}$

$$k_{t+1} / z_t k_t^{\alpha} - k_{t+2} / z_{t+1} k_{t+1}^{\alpha} * k_{t+1} / z_t k_t^{\alpha} = \beta \, \alpha \, - \beta \, \alpha \, k_{t+1} / z_t k_t^{\alpha}$$

Let $x_t = k_{t+1} / (z_t k_t^{\alpha})$ the savings rate

$$x_t - x_{t+1} * x_t = \beta \, \alpha \, - \beta \, \alpha \, x_t$$

Solve for forward mapping:

$$x_{t+1} = 1 + \beta\alpha - \frac{\beta\alpha}{x_t}$$

Since we are solving an infinite horizon problem the steady state level of savings occurs when $x_{t+1} = x_t$

$$x_t^2 - (1 - \beta\alpha)x_t + \alpha\beta = 0$$

Use quadratic formula to find that

$$x_t = \alpha\beta$$

So the policy function on savings would be

$$k_{t+1} = \alpha\beta z_t k_t^{\alpha}$$

and plug back into the budget constraint to find c_t.